Explainable AI - Banana Problem Report

Student Details

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Subject: Explainable AI

1. Dataset Overview

The dataset 'disease_risk.csv' contains health metrics for 300 individuals. It consists of three clinical features: Age, BMI, and Systolic Blood Pressure. The target variable is 'disease', which indicates the presence (1) or absence (0) of a condition. The dataset is perfectly balanced with 150 instances for each class.

2. Model Performance

A Generalized Additive Model (GAM) using Logistic regression was trained on 80% of the data. The model uses smoothing splines (s-terms) for all continuous features to capture non-linear relationships.

Model Accuracy on Test Set: 60.00%

3. Selected Patient Explanation

Patient #266 was selected for analysis as the model correctly predicted 'Disease' for this instance.

- Age: 53.0
- BMI: 29.7
- Systolic BP: 145.4

Predicted Probability: 73.48%

Actual Label: Disease

4. Feature Contribution Analysis

The contribution of each feature to the log-odds of the prediction was calculated using the partial dependence functions of the GAM:

- Age contribution: 0.0778 (Increases risk)
- BMI contribution: 0.2180 (Increases risk)
- Systolic BP contribution: 0.2353 (Increases risk)
- Intercept/Bias: 0.4878

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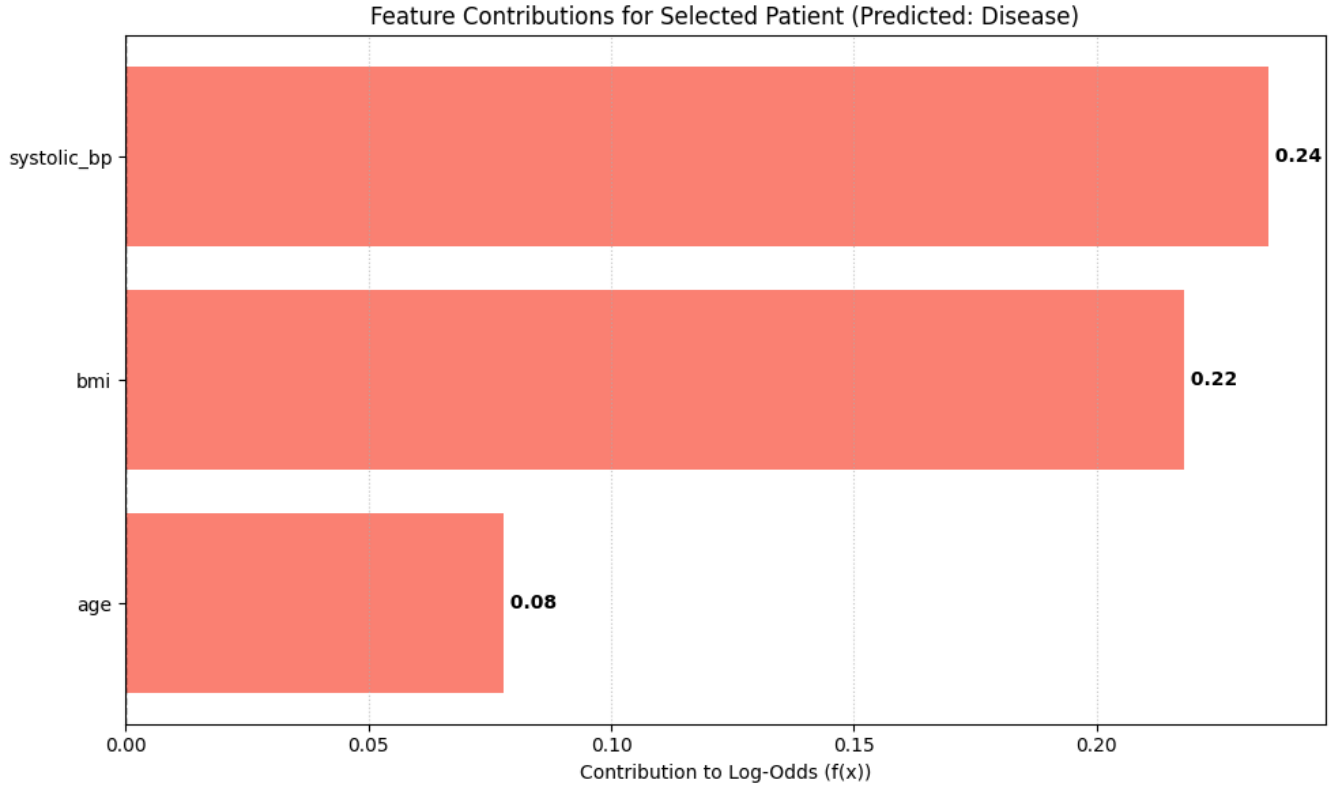
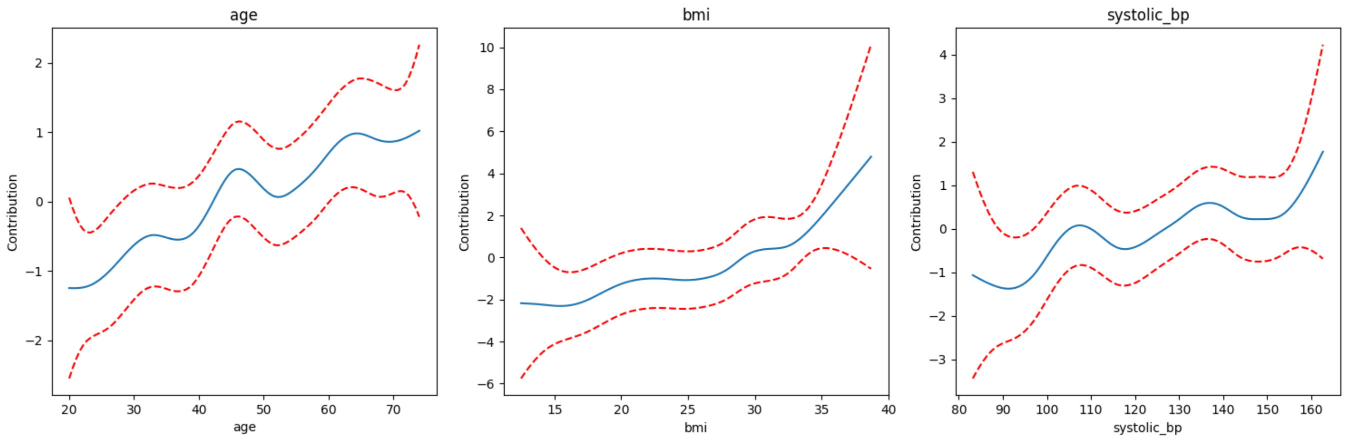
5. Visualizations

Local Explanation: Feature Contributions for Patient #266



Global Explanation: Feature Shape Functions



6. Summary

The analysis demonstrates that the GAM provides high interpretability by allowing us to see the exact numerical contribution of each feature. For Patient #266, Systolic BP was the most significant factor pushing the prediction towards 'Disease'. All features for this specific patient had a positive impact on the risk score, justifying the model's high-confidence prediction.